SCORE INTEGRITY UNDER MISSING MODALITIES: A RENORMALIZING MULTIMODAL SCORING ARCHITECTURE FOR AUTOMATED VIVA ASSESSMENT

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Viva voce assessment combines verbal explanation, delivery and interaction, so multimodal assessment requires consistent rules for using video, audio and transcript evidence. This thesis examines the Gradex Viva Evaluation scoring engine and its availability-aware aggregation policy. It formalises three behavioural families - visual engagement, acoustic delivery and transcript fluency - using bounded deterministic mappings, available-feature family means and renormalised family weights. A versioned feature schema keeps official scoring inputs distinct from supporting diagnostic and advisory services, while capture-quality checks and explicit output modes organise the evaluation pathway. The supplied research paper reports an evaluation using 20,000 sampled feature vectors, missing-data ablations, +/-10% input perturbations, parameter sweeps, correlated-degradation sampling and behavioural invariance tests. Its reported results show mean absolute score disturbance decreasing from 9.12 to 3.04 points on a 100-point scale and average grade-band changes decreasing from 65.8% to 29.5% across thirteen missingness conditions. These findings support availability-aware scoring as an interpretable software-design approach under the reported simulation conditions. The thesis connects the scoring policy to the wider Gradex implementation, including the browser client, media processing, transcript analysis, advisory LLM services and examiner-controlled technical assessment, and outlines a pathway toward further evaluation with real viva recordings and examiner rubrics.

Keywords: automated viva assessment, multimodal scoring, missing-data robustness, score integrity, educational technology

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional neural network
DSR	Design science research
HNR	Harmonics-to-noise ratio
LIME	Local Interpretable Model-agnostic Explanations
LLM	Large language model
LSTM	Long short-term memory
MFCC	Mel-frequency cepstral coefficient
NIST	National Institute of Standards and Technology
NLP	Natural language processing
RMS	Root mean square
SHAP	SHapley Additive exPlanations
WPM	Words per minute

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1 INTRODUCTION

1.1 Overall project context

Viva voce examinations allow examiners to probe reasoning, ask follow-up questions and observe how candidates explain decisions in real time. Unlike written tests, examiners can clarify ambiguity and judge answer structure. Oral examinations are difficult to scale consistently, and judgement may be affected by fatigue, context and assessment order [1]. These limits motivate support tools that complement rather than replace human judgement.

The group proposal, “AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System”, frames the wider Gradex platform through video, audio and transcript processing, including computer vision, speech-to-text, acoustic analysis and natural-language methods. The implementation inventory discussed in Chapter 5 describes the system layers and workflows relevant to the individual scoring component, while the research paper focuses on deterministic aggregation and score integrity as measurement availability varies.

1.2 Individual component context

The component examined in this thesis is the deterministic, renormalizing multimodal score-integrity engine described in “Score Integrity Under Missing Modalities: A Renormalizing Multimodal Scoring Architecture for Automated Viva Assessment”. In the Gradex platform, the implementation is identified as VivaEvaluationEngine, version 1.1. Its official score combines three behavioural families: visual engagement, acoustic delivery and transcript fluency. Each family contains bounded sub-features, and its score is calculated from the permitted measurements available for that family.

The individual contribution focuses on how the scoring policy handles varying measurement availability and separates official score inputs from supporting analysis. A deterministic scoring core, explicit capture-quality checks and defined output modes support a traceable calculation, with advisory language-model services kept separate from the official behavioural score.

1.3 Problem statement

Reliable multimodal viva assessment requires a consistent way to combine video, audio and transcript evidence across recording conditions. The scoring policy addresses this through availability-aware aggregation: each measured sub-feature is mapped to a bounded value, family scores average the eligible measurements, and available family weights are renormalised before computing the overall behavioural score. This preserves the distinction between a measured low value and a measurement that could not be produced, supporting consistent interpretation of the available evidence.

A defined feature-eligibility boundary complements the aggregation policy. The official behavioural score uses a versioned set of permitted inputs, while advisory LLM services, diagnostic signals and rubric-based technical accuracy operate in separate modules. Technical accuracy can be included through the explicitly configured WITH_TECHNICAL_ACCURACY mode, preserving the distinction between delivery-focused indicators and subject-matter assessment.

1.4 Research gap

Research on multimodal machine learning recognises fusion and missing modalities as core challenges [4], and recent work has proposed learned fusion architectures that remain useful when some medical modalities are absent [5]. Those approaches target predictive performance in their respective domains. An educational scoring system presents an additional governance requirement: the treatment of a missing signal itself changes an individual score and grade band. A learned reconstruction or default value may be technically reasonable but still needs validation before it can affect an academic outcome.

The individual paper contributes a focused engineering policy: a defined method for handling unavailable sub-features and complete feature families, together with a boundary around outputs that are permitted to affect the official mark. Its experiments compare the score behaviour of zero-imputation and availability-aware renormalisation under controlled missingness and perturbation conditions. These results provide evidence about the formula under the selected simulation design;

human-examiner agreement and performance across real recording conditions are separate validation questions addressed later in the thesis.

1.5 Aim and objectives

The aim is to analyse and document a deterministic multimodal scoring policy that preserves score integrity under missing measurements while making the boundary between permitted and unvalidated signals explicit.

Objective 1: Formalise bounded normalisation for the permitted engagement, acoustic-delivery and transcript-fluency sub-features.

Objective 2: Define how available sub-features form family scores and how unavailable sub-features or families are omitted and reweighted.

Objective 3: Define capture-quality checks and the input policy that governs which signals can contribute to the official score.

Objective 4: Evaluate the scoring function’s response to missing-data ablations, local input perturbations, unfitted-constant sweeps and correlated degradation using the quantitative results reported by the supplied research paper.

Objective 5: Identify the limitations and evidence needed before interpreting the behavioural score as valid support for human viva assessment. Table 1.1 maps these questions to the available evidence and the claims it cannot establish.

Table 1.1. Research questions and evidence sources

Research question	Evidence source	What it can establish	Interpretation / next step
RQ1: missing-data impact	Reported 20,000-vector ablation	Score and grade disturbance under selected ranges	Synthetic results; real-viva performance is a complementary evaluation step
RQ2: sensitivity	Reported ±10% perturbations and parameter sweeps	Local stability and ranking consistency in the sampler	Parameter stability is evaluated within the tested ranges
RQ3: correlated loss	Second simulated population	Effect of a specified dependency scenario	Recording-quality dependencies can be measured in the next study
RQ4: assessment validity	No human labels supplied; future protocol in Chapter 6	Validation requirements	Examiner agreement and subgroup outcomes can be examined with rubric data

1.6 Research questions or hypotheses

RQ1: How does dynamic renormalization change the score disturbance and grade-band changes caused by unavailable sub-features or complete feature families compared with zero-imputation?

RQ2: How stable is the scoring function when raw inputs are perturbed and when engineering constants are varied within the ranges reported in the research paper?

RQ3: Does the reported advantage of renormalization remain when missingness is correlated across signals in a simulated recording-quality scenario?

RQ4: What evidence is required before the resulting behavioural score can be used as a defensible component of academic assessment?

These questions separate software-behaviour evidence from construct validation. The quantitative experiments can answer the first three within the sampler and scoring implementation described by the paper. The fourth cannot be answered by a synthetic simulation alone; it requires human-examiner ratings, real recordings, subgroup analyses and a documented governance process.

1.7 Scope limitations and assumptions

The scope is the scoring core and its availability-aware policy. It covers the three defined feature families, bounded normalisers and thresholds described in the paper, Mode A and Mode B, the feature-eligibility boundary, capture-quality states and the reported robustness experiments. The wider Gradex architecture is described where it

clarifies how official scoring interacts with video, audio, transcript, storage and advisory services.

The quantitative findings are discussed as reported in the supplied research paper and interpreted within its controlled simulation design. The scoring output is positioned as a behavioural and delivery measure intended to support examiner review alongside technical assessment. A complementary study using authentic viva recordings, a written rubric and qualified human raters can extend the evaluation to examiner agreement and construct alignment.

1.8 Individual contribution

The individual contribution comprises four parts: framing missingness as a scoring-policy issue rather than a low feature value; specifying availability-aware family aggregation and weight renormalization; defining an exclusion boundary and feature-status ledger; and analysing paper-reported score disturbance, grade-band changes, input sensitivity, parameter sensitivity and correlated degradation.

The individual contribution documented here is the formalisation and analysis of the scoring policy, its permitted-input boundary, reported robustness results and validation requirements. Shared implementation and testing work should be allocated according to the project's agreed individual-contribution record so that the thesis distinguishes the candidate's analytical contribution from wider team deliverables.

1.9 Thesis structure

Chapter 1 frames the problem; Chapter 2 reviews related work; Chapter 3 sets out methodology; Chapters 4–5 describe design, implementation and testing; Chapters 6–7 report and interpret results; Chapter 8 covers commercialisation and sustainability; Chapter 9 concludes with objective achievement and future validation.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review is a focused technical synthesis rather than a systematic review with a registered search protocol. Its primary scope follows the supplied research paper and proposal: multimodal fusion under missingness, behavioural signals in oral assessment, speech transcription, facial landmark analysis, AI-assisted viva systems and methods for explaining machine-learning outputs. Bibliographic details were checked against accessible publishers, scholarly-index and institutional records where possible. The review avoids using the proposal's unverified numerical accuracy claims as evidence for the individual component.

The review uses three levels of relevance. Directly relevant work concerns missing modalities or score behaviour. Adjacent work concerns the use of speech, visual and linguistic signals in assessment or interview settings. Governance-oriented sources inform how to treat unvalidated or high-impact signals. The key analytical distinction is between showing that a modality can be processed and demonstrating that a specific derived feature validly measures the construct for which a student is being assessed.

2.2 Domain and theoretical foundations

Multimodal learning combines evidence from different input types, such as video, sound and text. Baltrušaitis, Ahuja and Morency describe fusion, representation, alignment and co-learning as recurring challenges in multimodal machine learning [4]. These challenges apply to viva analysis because visual, acoustic and transcript streams are generated at different rates, have different failure modes and may not represent the same aspect of performance. An integrated system therefore needs rules not only for how to combine valid signals, but also for what to do when a stream is unusable.

A low score derived from the harmonics-to-noise ratio (HNR) means that the signal was measured and mapped to a low value; a missing HNR measurement means that the estimator did not supply a usable value. The two states carry different evidence. Zero-imputation collapses them into the same numeric representation, while availability-aware scoring preserves that difference. The distinction matters in assessment because the score is interpreted as a statement about a person even when a zero may actually reflect a recording or processing condition.

Renormalisation is a deterministic aggregation policy. If only a subset of a family's sub-features is available, the family score is the arithmetic mean of those available values. If a whole family is unavailable, its nominal weight is removed from the denominator and the remaining available family weights are rescaled to sum to one. The policy preserves the distinction between observed feature values and availability metadata. The configured assessment status is recorded alongside the score pathway when the available input coverage requires review.

Construct validity is an important foundation. Engagement, vocal fluency and delivery describe behavioural aspects of a viva; technical correctness is handled as a separate assessment dimension. The scoring paper therefore keeps an optional technical-accuracy score separate from behavioural delivery indicators, reflecting the measurement principle that distinct constructs should be clearly defined before they are combined.

2.3 Related research

Work on missing modalities offers a useful point of comparison. Wang et al. present a tri-modal fusion architecture for medical data and train it to remain useful when image, text or structured data may be unavailable [5]. That work uses learned fusion and multivariate losses in clinical prediction tasks. It illustrates an alternative to deterministic availability-aware scoring and supports treating modality availability as an explicit design condition. The Viva Evaluation scoring engine uses a simpler policy that omits unavailable measurements and renormalises the remaining permitted family weights.

Teacher grading of open responses can be affected by contextual factors and fatigue [1], motivating consistent procedures but not proving automation is more accurate. Tao et al. study confidence estimation from behaviour in immersive learning [2], and Reddy et al. are cited for audio analysis of confidence in viva responses [3]. Transfer across contexts depends on sample, language, recording conditions and the construct definition; availability alone is not external validity.

Khalil et al. review deep-learning methods and datasets for speech emotion recognition and discuss its challenges [6]. The scoring engine uses selected acoustic delivery features, while emotion labels are retained as supporting diagnostic outputs. This

separation keeps the official score aligned with its defined delivery construct and allows emotion analysis to be considered separately for context-specific applications.

MediaPipe supports perception pipelines and is used in the wider project for face landmarks [7]. Ben et al. survey video-based micro-expression analysis [8]. Head pose, gaze, blinking and facial movement can be affected by disability, culture, camera placement, lighting and communication style; their use in grading therefore requires construct-validity and subgroup evidence.

Speech transcription supplies text for downstream features. Radford et al. describe Whisper, a speech-recognition system trained through large-scale weak supervision [9]. Its availability makes transcript-derived measures feasible, but the transcript remains a model output with possible recognition errors, including variations associated with accent, domain vocabulary and recording quality. Bidirectional Encoder Representations from Transformers (BERT) introduced contextual language representations and has enabled a range of text-understanding tasks [10]. Such models may support future rubric-grounded content analysis, but the official score deliberately does not treat large language model (LLM) evaluation or semantic criterion scores as official behavioural inputs.

Senarathne et al. describe a natural language processing (NLP)-based interview automation system [11]. Cao and Zahid discuss generative AI for viva voce and coursework authentication [12], while Ipeirotis and Rizakos describe voice-AI oral assessment and transcript grading [13]. These systems expand prompting, conversation control and content assessment. The contribution here complements those services by defining a consistent scoring policy for the feature signals available during an assessment.

Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) support interpretation of model predictions [14], [15]. The scoring design complements such approaches with score provenance: each official input belongs to a named family with a recorded role, and behavioural tests cover the separation of supporting outputs from the mark. Provenance makes the scoring pathway easier to review and can be combined with attribution methods where appropriate.

2.4 Existing systems products and techniques

Four missing data techniques are especially relevant. The first is zero-imputation, where missing values are replaced with numeric zero. It is simple and deterministic, but it confuses unavailable with poor and can reduce a score systematically. The second is mean or model-based imputation, which attempts to infer a plausible value from other observations. This can be useful in predictive modelling, but it adds assumptions and possibly learned behaviors that require validation. In a high-stakes assessment, an imputed value should be visible as estimated rather than silently treated as observed.

The third technique is complete-case handling, which requires every expected input before scoring. The fourth is availability-aware renormalisation, adopted by the Viva Evaluation scoring engine. It calculates family means from permitted measurements and adjusts family weights to the available evidence, maintaining a deterministic and traceable calculation when input availability varies. Capture-quality checks and configured output states complement this policy within the complete assessment workflow.

The wider proposal and implementation inventory cover convolutional neural network (CNN) and long short-term memory (LSTM) models, face-analysis libraries, acoustic feature extraction, transcription, language-model services and explainability. Within the official behavioural score, the permitted families are visual engagement, acoustic delivery and transcript fluency; diagnostic outputs such as raw gaze/blink measures, RMS energy, MFCCs and emotion labels remain available for supporting analysis rather than being automatically combined into the mark. Table 2.1 compares the relevant technique families and their implications for this design.

Table 2.1. Related-work synthesis and implications

Research area	Representative source	Useful contribution	Implication for viva scoring
Multimodal learning	Baltrušaitis et al. [4]	Organises fusion, alignment and representation challenges	Provides a foundation for designing assessment-specific availability rules
Missing modalities	Wang et al. [5]	Demonstrates trained robustness to absent modalities in medical tasks	Illustrates a complementary domain and learned-fusion approach
Speech/emotion	Khalil et al. [6]; Tao et al. [2]	Motivates analysis of acoustic and behavioural features	Supports context-aware interpretation of behavioural signals
Transcription and language	Radford et al. [9]; Devlin et al. [10]	Enables transcript-derived measures and semantic processing	Motivates transcript-quality checks and careful downstream use
Automated oral assessment	Cao and Zahid [12]; Ipeirotis and Rizakos [13]	Expands questioning and transcript grading capabilities	Complements availability-aware scoring with interaction and content assessment
Interpretability	Ribeiro et al. [14]; Lundberg and Lee [15]	Provides methods for explaining model outputs	Complements input governance through model-attribution methods

2.5 Gap synthesis and conceptual framework

Prior literature motivates multimodal evidence while highlighting the importance of defining the intended assessment construct and its scoring policy. The contribution of the Viva Evaluation scoring engine is a deterministic, auditable mechanism for handling availability and controlling which signals enter the official score. This engineering layer complements the wider work of construct definition, rubric design and stakeholder evaluation.

The conceptual framework has four layers: acquisition and quality checks; permitted feature extraction and bounded normalisation; availability-aware aggregation and assessment-status generation; and evaluation with human oversight. Acquisition distinguishes capture conditions from measured feature values, while scoring applies deterministic and bounded transformations. The simulations evaluate software behaviour under missingness and perturbation, and the framework supports continued study of construct alignment, examiner agreement, accessibility and practical usefulness.

3 RESEARCH METHODOLOGY

3.1 Research design

The group proposal describes a Design Science Research (DSR) approach for a real-world viva-assessment problem, moving through problem identification, objectives, design and development, demonstration, evaluation and communication. The individual component applies an implementation-centred quantitative evaluation to a deterministic scoring function. These descriptions operate at different levels: DSR provides the larger project context, while the reported experiments quantify how the scoring policy behaves under controlled inputs and simulated failure conditions.

The evaluation applies a controlled software-behaviour design rather than model training. It compares availability-aware renormalisation with zero-imputation and measures score delta, grade-band change, sensitivity to local input perturbations, parameter sweeps and correlated degradation. The design makes it possible to inspect how the calculation behaves under specified inputs and missingness patterns; examiner-led evaluation is the next step for connecting these engineering findings to assessment practice.

3.2 Data participants or study materials

The evaluation dataset is a synthetic population of 20,000 feature vectors described in the supplied paper, sampled over controlled ranges for continuous features and count distributions for transcript indicators. This setup supports a controlled comparison of scoring policies under defined missingness conditions. The study therefore focuses on score-policy behaviour, with recorded viva sessions and examiner ratings forming a complementary route for further evaluation.

The baseline continuous ranges are CNN engagement and facial confidence from 0.30 to 0.95; pitch standard deviation from 10 to 140 Hz; HNR from 5 to 32 dB; jitter from 0.002 to 0.040; shimmer from 0.020 to 0.180; speech rate from 80 to 200 words per minute; and sentence completion from 0.50 to 1.00. Hedge, filler and long-pause counts are sampled from Poisson distributions with means 3, 5 and 2. A second population uses latent recording-quality and speaker-fluency factors to model correlated degradation. Table 3.1 records the ranges and distributions reported by the

paper; a repeatable release package can preserve the corresponding generator, seed and aggregate outputs alongside its results.

Table 3.1. Feature ranges used in the reported simulation

Input	Sampling range / distribution	Role in scoring
Convolutional neural network (CNN) engagement, facial confidence	Uniform [0.30, 0.95]	Engagement family
Pitch standard deviation	Uniform [10, 140] Hz	Pitch stability normalizer
HNR	Uniform [5, 32] dB	harmonics-to-noise ratio (HNR) clarity normalizer
Jitter / shimmer	Uniform [0.002, 0.040] / [0.020, 0.180]	Articulation normalizer
Speech rate	Uniform [80, 200] words per minute (WPM)	Transcript rate band
Sentence completion	Uniform [0.50, 1.00]	Transcript family
Hedges / fillers / long pauses	Poisson means 3 / 5 / 2	Transcript family count scores

3.3 Experimental or study procedure

The reported procedure begins with a canonical input vector whose feature values and availability statuses are passed to the scoring function. Each raw value is mapped into a bounded score in the interval [0, 1]. For each family, the system averages only its available sub-features; if the family has no available sub-feature, that family is unavailable. The official score is the weighted mean of the available family scores after the available family weights are normalized to sum to one. The alternative baseline retains the nominal set of sub-features and assigns zero to unavailable values.

Five experiment groups are reported. Experiment 1 ablates individual features and complete families under both missing-data policies, recording mean signed score change, mean absolute change, grade-band changes and rank correlation. Experiment 2 perturbs each raw input by plus or minus 10 percent while the other inputs remain fixed. Experiment 3 sweeps each unfitted normaliser constant and speech-rate band. Experiment 4 repeats the comparison on a second population with correlated degradation. Experiment 5 covers behavioural tests, including invariance of the official score to changes in LLM criterion scores and diagnostic aggregates. The project inventory records 189 passing Python unittest tests as of 30 August 2026; the supplied paper separately describes a verification pass in September 2026. Figure 3.1 summarises the reported experiment and evidence workflow.

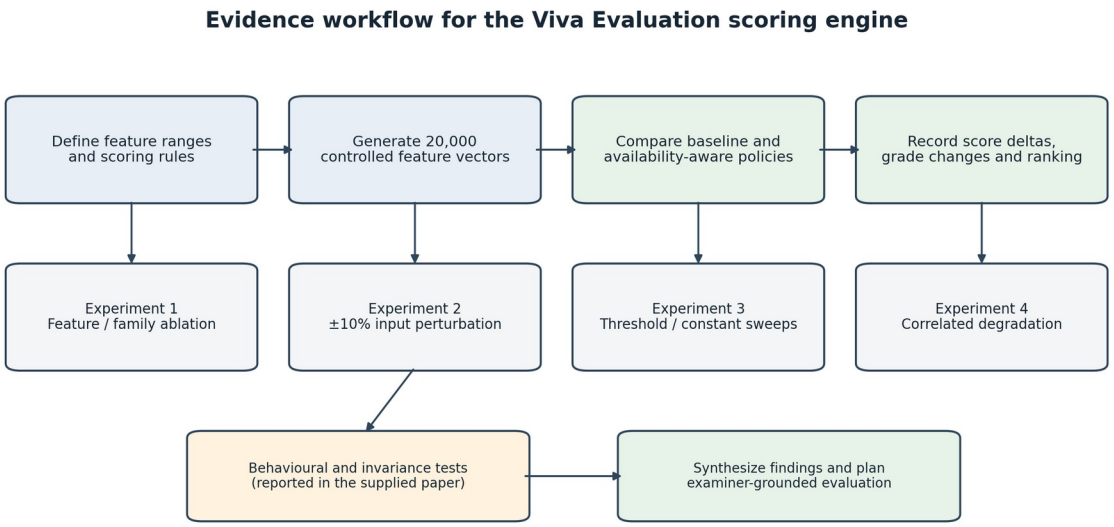


Figure 3.1. Evidence workflow for the Viva Evaluation scoring engine

3.4 Measures and evaluation metrics

Mean signed score delta captures direction: a negative value indicates that a missingness condition lowers the mean score relative to the complete vector; a positive value means the remaining families receive greater normalized weight and produce a higher average for the specified sampler. Mean absolute score delta measures the magnitude of disturbance without cancellation between positive and negative cases. Grade-band change rate indicates the proportion of candidates moved into a different letter-grade band by the missingness policy. Spearman rank correlation quantifies preservation of candidate ordering after a feature or family is removed.

Perturbation sensitivity is assessed from the absolute score movement caused by a 10 percent change in input. Constant sensitivity is assessed through mean and maximum score change and rank correlation between the baseline and each parameter sweep. These metrics answer different questions. Low score disturbance supports calibration stability under the tested failure condition, whereas high rank correlation supports ordering stability. The supplied paper notes a trade-off: zero-imputation preserves rank marginally better on average in its sampled population but disturbs absolute scores more, which is relevant when fixed grade thresholds are used.

3.5 Data analysis

The baseline synthetic population reported by the paper has a mean score of 59.28 and a standard deviation of 7.36. These values describe the controlled sampling

distribution and provide a reference point for interpreting score changes across the experiments. The missing data analysis pools thirteen conditions, including individual sub-features and the three whole families. The correlated-degradation population is wider, with reported standard deviation 13.40 and correlation of +0.71 between HNR and facial confidence as one example of dependency. These distributional properties reflect the generator assumptions and should not be interpreted as observed recording-quality relationships.

The numerical findings are presented consistently across the tables and narrative, with interpretation focused on the observed pattern over the defined simulation conditions. The deterministic scoring function and the specified sample provide a comparative view of policy behaviour over the selected synthetic input space. The reported ranges, distributions and thresholds define the study conditions; a recording-based follow-up can inform configuration for institution-specific assessment practice.

3.6 Ethics privacy security and responsible AI

No human participant data were supplied or analysed for the reported experiments. This reduces direct privacy risks within the simulation but does not remove the ethical stakes of the intended application. Real viva recordings contain identifiable facial, voice and educational-performance data. Any future empirical study should use institutionally approved protocols, informed consent, access control, minimum retention, secure storage and defined deletion procedures. Where recordings are used for validation, the system should retain a separation between raw recordings, extracted features and score outputs, and should restrict access to each according to role.

Responsible-AI governance complements the scoring policy. Recording environment and communication context can shape the available signals, so evaluation should consider accessibility, accent, camera quality, connectivity and other relevant conditions. NIST’s AI Risk Management Framework and ISO/IEC 23894 provide lifecycle-oriented reference points [16], [17], while UNESCO guidance emphasises human-centred and privacy-conscious use of generative AI in education [18]. Calibration, subgroup review and a human review pathway support responsible use across different institutional settings.

3.7 Reproducibility and research artefacts

The reported evaluation uses a seeded simulation procedure and the scoring function described in the supplied paper. Across 20,000 feature vectors and the defined experiment groups, the analysis compares availability-aware aggregation with zero-imputation using score deltas, grade-band movement, perturbation sensitivity, parameter sweeps and correlated degradation. The findings are attributed to the published study and provide a structured reference for subsequent evaluation using representative recordings and examiner rubrics.

A reproducibility package can preserve the scoring-engine version, dependency lockfile, random seed, generator, aggregate outputs, test-runner summary and feature-status ledger alongside released results. Recording these artefacts together supports consistent review of the scoring rules across software versions and makes future calibration studies easier to reproduce.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements translate the research paper's design commitments into testable engineering criteria. They focus on deterministic scoring, bounded mappings, explicit availability, separation of behavioural and technical content, defined assessment statuses and a testable input boundary. These criteria provide a basis for acceptance testing and release-level review.

4.2 Component architecture within the complete solution

The broader proposal describes a pipeline beginning with a recorded viva and branching into video analysis, audio analysis and speech-to-text/content processing. The video branch proposes frame extraction, face detection, facial-expression analysis and video-derived confidence features. The audio branch proposes preprocessing, voice feature extraction and speech-emotion processing. The text branch proposes transcription, cleaning and semantic analysis. In the later paper's description of the implementation, scoring engine receives a narrower set of permitted values: CNN mean engagement and facial confidence, pitch stability, HNR clarity, jitter/shimmer articulation and five transcript measures.

The scoring engine sits between feature extraction and reporting. A versioned canonical feature schema defines eligible inputs, while the feature-status ledger connects measured values to their scoring roles. Capture-quality checks cover face quality, multi-face conditions and lip-motion consistency with transcribed speech. The resulting assessment metadata accompanies the score, and diagnostic outputs and language-model services remain separate from the official behavioural mark. Mode B combines the behavioural score with examiner-entered technical accuracy.

Figure 4.1 presents the scoring architecture, while Table 4.1 records the permitted feature families and the roles of supporting diagnostic inputs. Together, they define the relationship between media processing, the scoring engine and examiner-facing outputs.

Table 4.1. Official-score feature families and input policy

Family/status	Permitted inputs or examples	Policy
Engagement	Mean CNN engagement; facial confidence	Average available values; one value is sufficient

Family/status	Permitted inputs or examples	Policy
Acoustic	Pitch stability; HNR clarity; jitter/shimmer articulation	Average available terms; voice-quality failure may remove HNR and articulation together
Transcript	Rate band; hedges; fillers; long pauses; completion	Average available terms; no model answer needed
Supporting analysis	Emotion labels; raw gaze; blink counts; RMS; mel-frequency cepstral coefficients (MFCCs)	Available for supporting analysis and experiment design; separated from official score inputs
Extension features	Temporal CNN-LSTM; response structure; pause totals; head-event detection	Candidates for future assessment-focused evaluation
Advisory / optional	LLM narrative; Q&A relevance; external technical accuracy	Advisory feedback remains separate; examiner-entered technical accuracy is combined in Mode B

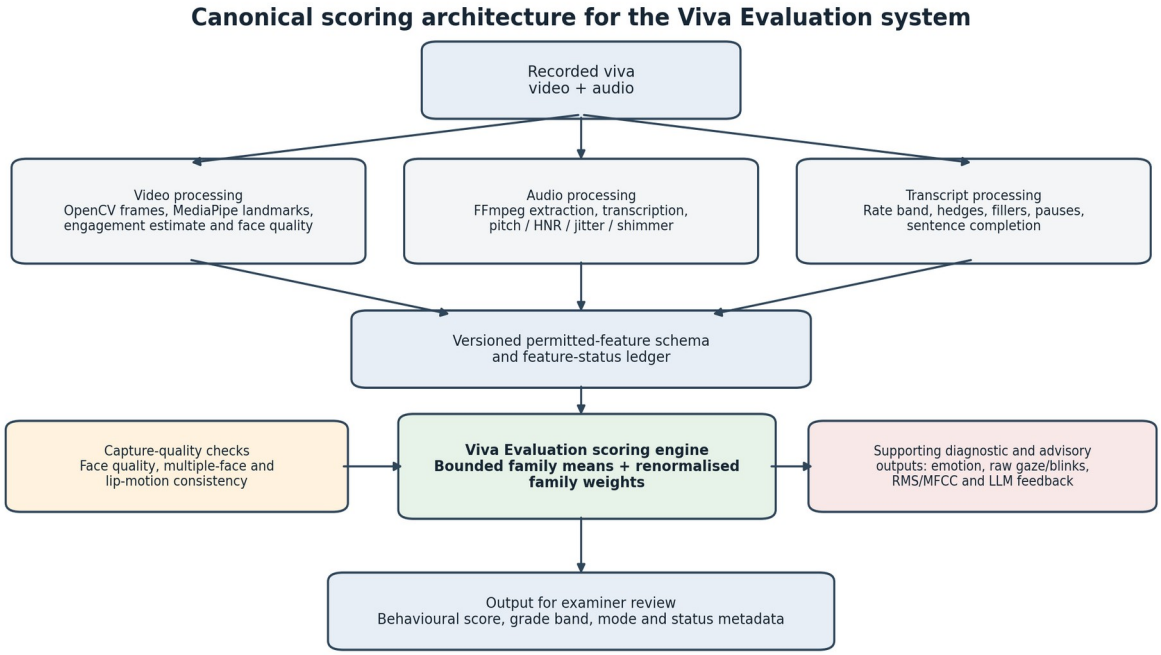


Figure 4.1. Canonical scoring architecture for the Viva Evaluation system

4.3 Design alternatives and decisions

Zero-imputation is computationally simple and produces a number even when data is missing. Its disadvantage is semantic: the numeric default represents a low feature value, not the absence of a measurement. The paper's results show that the distinction matters at both score and grade level. A second alternative, imputing a likely value

from other modalities, would require a fitted model or rule and a validation study that shows the imputed signal is appropriate for the intended assessment context.

Complete-case rejection requires every expected input and can discard otherwise usable evidence. The selected policy instead calculates from permitted measurements, renormalises family weights over the available families, and records the configured assessment status. This keeps the rule deterministic and makes the handling of input availability explicit in the architecture and test plan.

Table 4.2. Functional requirements and acceptance criteria

ID	Requirement	Acceptance evidence
FR-01	Deterministic score for identical permitted input and availability mask	Locked fixture returns the same score/status on re-analysis
FR-02	Bounded normalizers in [0, 1]	Boundary tests cover below/inside/above thresholds
FR-03	Availability metadata accompanies measured features	Family means use eligible observed values
FR-04	Dynamic weight renormalization	Available family weights sum to one
FR-05	Capture-quality checks govern score eligibility	Gate outcomes are recorded with assessment metadata
FR-06	Supporting signals remain separate from the official mark	Invariance tests cover advisory and diagnostic outputs
FR-07	Mode B incorporates examiner-entered technical accuracy	The configured mode records the score combination and publication status
FR-08	Feature status is auditable	Ledger records measurement, scoring role and input provenance

4.4 Detailed design

Let A_i be the set of available sub-features for family i. The family score is the mean of the available values: $E_i = (1 / |A_i|) \times \Sigma(k \text{ in } A_i)\ s_k$, when A_i is non-empty. If no sub-feature is available, E_i is undefined and the family is omitted. The set F contains the available families, and each available family has nominal weight $w_i = 1$. The official behavioural score is $S_{AI} = 100 \times \Sigma(i \text{ in } F)\ \tilde{w}_i E_i$, where $\tilde{w}_i = w_i / \Sigma(j \text{ in } F)\ w_j$. Therefore the normalized family weights sum to one for any non-empty F.

The engagement family averages the mean CNN engagement estimate and normalized facial confidence when both are present; when only one is available, the available value forms the family score. Acoustic inputs comprise pitch stability, HNR clarity and articulation derived from jitter and shimmer. Pitch stability is represented as $s_p = 1 - \text{clamp}(\sigma_f / 120, 0, 1)$; HNR clarity is $s_{hnr} = \text{clamp}(HNR / 30, 0, 1)$. A lower-is-

better voice-quality metric x is mapped through q(x) = clamp (x_poor − x) / (x_poor − x_good), 0, 1). Articulation is s_a = 0.5 q(jitter) + 0.5 q(shimmer), with configured good/poor thresholds of 0.005/0.030 for jitter and 0.030/0.150 for shimmer. The paper identifies these values as engineering choices, not fitted parameters.

The transcript family contains five deterministic measures: speech-rate band, hedge score, filler score, long-pause score and sentence completion. The rate score is 1.00 inside the configured 120–160 words-per-minute band and 0.65 outside the band. The other measures use count-based normalisers: s_hed = 1 − min(1, N_h / 8), s_fill = 1 − min(1, N_f / 12), and s_long-pause = 1 − min(1, N_l / 6), with N_l counting pauses greater than two seconds. Sentence completion is clamped to [0, 1]. These thresholds and divisors are configurable engineering parameters that can be calibrated against an agreed assessment rubric.

The scoring design provides two output modes. Mode A returns the behavioural score. Mode B combines the behavioural score with an examiner-entered technical-accuracy score T in [0, 10], using equal weight: S_final = 0.5 S_AI + 0.5 (T / 10 × 100), after examiner publication. Grade bands are A+ (90–100), A (80–89), B+ (70–79), B (60–69), C+ (50–59), C (40–49) and F (below 40). The paper reports a configured fixture result of 73.76 (B+).

Figure 4.2 outlines how the scoring workflow moves from input availability and capture checks through family aggregation to the configured assessment mode.

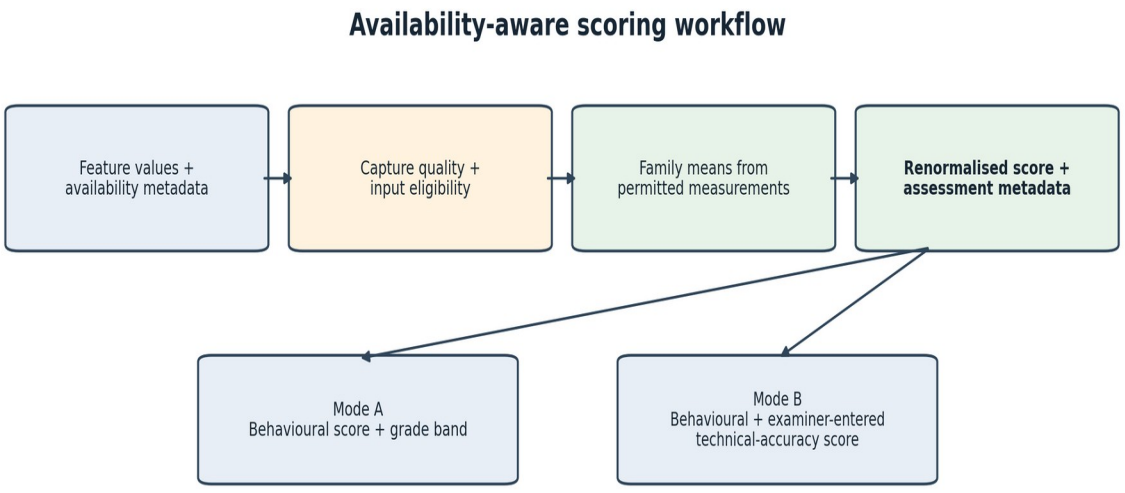


Figure 4.2. Availability-aware scoring workflow

4.5 Standards and professional practice

The design follows several general software-quality principles: explicit input schemas, bounded values, deterministic transformations, clear status states, isolated modules and invariant-based tests. These practices make the calculation auditable without overclaiming full model explainability. A ledger should distinguish at minimum: measured, normalised, scored, diagnostic, excluded and awaiting validation. Each entry should identify the provenance of a signal, the version of the extraction process, the range mapping and the reason it is eligible or ineligible for scoring.

The university formatting standards specify Times New Roman body text at 12 pt, 1.5-line spacing, 40 mm left and bottom margins and 25 mm top and right margins; they also require tables and figures to be cited in the text. In the system, professional practice is supported by a documented rubric, human oversight, clear communication about what the score represents, a review pathway and defined handling of sessions requiring further assessment. NIST AI RMF and ISO/IEC 23894 provide lifecycle-oriented reference points for ongoing governance and compliance planning.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The Gradex Viva Evaluation platform uses Python 3.10 (the project inventory notes compatibility across Python 3.10-3.12), FastAPI/Uvicorn, Pydantic and httpx for validated REST and live WebSocket services. MongoDB persistence uses PyMongo/Motor, with published marks stored in vivamark.marks. The React 18/TypeScript examiner client uses Vite, Tailwind CSS 4, MUI and Recharts; FFmpeg supports media extraction. The project inventory dated 30 August 2026 records 189 passing Python unittest tests.

Video processing uses OpenCV and MediaPipe Face Landmarker for frame analysis and 478 facial landmarks, including iris points 468-477. PyTorch, torchvision and timm support EfficientNet-B0 inference: engagement_cnn.pt provides the frame-level engagement input for the official mark, while hsemotion_improved.pt with enet_b0_8_best_afew.pt supports per-frame emotion diagnostics. The landmark pipeline derives gaze estimates (configured threshold 0.04), head-pose/stability proxies and Eye Aspect Ratio blink measures sampled at approximately 10 fps. Separate checks cover face quality, multiple-face detection and lip-motion consistency. A temporal CNN-LSTM extension point is reserved for future model integration.

Table 5.1. Viva Evaluation technology stack and module boundaries

Technology / layer	Role in the Viva Evaluation system	Scoring boundary / status
MongoDB; PyMongo; Motor	Persistence for evaluation records and published marks (vivamark.marks)	Database layer for results and review records
React 18; TypeScript; Vite; Tailwind CSS 4; MUI; Recharts	Browser client, examiner workflow and visual reporting	Presentation layer for assessment review and results
OpenCV; FFmpeg; MediaPipe Face Landmarker (478 landmarks; iris points 468-477)	Frame handling, audio extraction, face quality, multi-face and lip-motion checks; gaze threshold 0.04; head-pose/stability proxies; blink EAR at about 10 fps	Provides structured visual and media features for the scoring workflow
PyTorch; torchvision; timm; EfficientNet-B0; engagement_cnn.pt; hsemotion_improved.pt; enet_b0_8_best_afew.pt	Frame-level engagement inference and per-frame facial-emotion analysis	engagement_cnn.pt supplies the official engagement-family input; emotion labels support diagnostic analysis

Technology / layer	Role in the Viva Evaluation system	Scoring boundary / status
Whisper small; whisper-large-v3-turbo; Praat-Parselmouth; librosa; soundfile; SciPy; NumPy; torchaudio	Local transcript with word timestamps; live transcription; pitch, jitter, shimmer, HNR, 13 MFCCs at 16 kHz and RMS	Mapped pitch stability, HNR clarity and articulation support the acoustic family; transcript timestamps support rate and pause features
Transformers superb/wav2vec2-base-superb-er; ser_worker.py; optional SpeechBrain; MFCC/k-means diarization; optional Pyannote 3.1	Speech-emotion inference in an isolated worker and student/lecturer separation	Supporting analysis is separated from the official behavioural mark
Rule-based transcript measures	Rate band (120-160 WPM), hedges, fillers, pauses over 2 seconds, sentence completion and fragmentation	Deterministic transcript family with configurable engineering ranges
Groq; openai/gpt-oss-20b; openai/gpt-oss-120b; Gemini gemini-2.0-flash; OpenRouter; browser Web Speech API	Follow-up prompts, narrative feedback, Q&A relevance, technical-accuracy assessment, live copilot and transcription fallback	LLM judge and Q&A relevance are advisory; technical accuracy is examiner-published in WITH_TECHNICAL_ACCURACY mode
Viva Evaluation scoring engine, version 1.1; Python unittest	100-point, three-family score; WITHOUT_TECHNICAL_ACCURACY and WITH_TECHNICAL_ACCURACY modes; project inventory reports 189 passing tests on 30 August 2026	Mode with technical accuracy combines the rubric score at 50% and remains a draft until examiner publication

5.2 Component implementation

Implementation separates feature extraction, advisory services and official score calculation through a versioned canonical schema. Video, audio and transcript modules supply the input measurements; the scoring engine reads the permitted fields, while the live interviewer copilot, question recommender, narrative LLM judge and Q&A relevance modules support examiner workflows. The live copilot combines browser Web Speech recognition with Groq Whisper (whisper-large-v3-turbo), streamed chat and pooled httpx requests through REST/WebSocket services. The question recommender proposes candidate follow-up prompts for examiner selection. The official 100-point score uses three equally weighted families: engagement, acoustic delivery and transcript fluency. Bounded normalisers map eligible measurements to [0, 1]; available sub-features are averaged within each family, and family weights are renormalised across measurable families. The acoustic family uses mapped pitch-stability, HNR and jitter/shimmer-derived articulation measures; transcript inputs include speech-rate band, hedges, fillers, pauses over two seconds and

sentence-completion measures. A parallel feature-complete baseline can be reported for analysis without changing the official mark. Groq models openai/gpt-oss-20b and gpt-oss-120b support follow-up questions, narrative feedback, Q&A relevance and technical-accuracy services, with configured Gemini gemini-2.0-flash and OpenRouter fallbacks. Narrative judge and Q&A relevance outputs are advisory; rubric-based technical accuracy contributes 50% only in WITH_TECHNICAL_ACCURACY mode and remains a draft until examiner publication.

5.3 Integration with the other components

Uploaded-recording and live-assistance workflows share the same scoring pathway after media preparation and feature extraction. The integration combines the React examiner client, REST/WebSocket services and MongoDB persistence with a dedicated SER worker and configured provider fallback for advisory language services. Published marks are stored through PyMongo/Motor in vivamark.marks. The architecture supports multimodal processing, an inspectable scoring rule, modular services and examiner control of the final technical-accuracy publication.

Transcript and conversation modules include rule-based speech-rate, filler, hedge, pause and sentence-completion measures, as well as Q&A relevance and narrative feedback. The official mark remains governed by the permitted feature schema; supporting LLM services add examiner context without directly changing the behavioural score.

5.4 Verification and software testing

The project implementation inventory records 189 passing Python unittest tests as of 30 August 2026. The supplied paper separately describes a test-suite re-verification on 1 September 2026 and identifies checks for grade boundaries, Mode A/Mode B behavior, score invariance to advisory LLM outputs and diagnostic aggregates, capture-integrity gates, and the no-scorable-family state. Together, these reports describe a testing approach centred on deterministic behavior and separation of scoring from advisory modules.

The described tests focus on repeatable scoring behaviour and module boundaries. Keeping the versioned test suite, fixtures and configuration alongside each release

supports regression checks whenever a normaliser, threshold or permitted input changes.

5.5 Deployment and operation

Operationally, the system supports uploaded recordings and live examiner assistance, with REST/WebSocket services, a React examiner client, MongoDB persistence, isolated SER processing and provider fallback for advisory language services. The modular structure supports maintainable integration and service resilience; runtime, service health and throughput can be monitored under representative workloads.

For responsible operation, present the behavioural score with its mode, contributing families, availability metadata and review status. Scoring versions, permitted inputs and manual overrides should be recorded, while media access and retention follow institutional policy. These controls support traceability, privacy-conscious operation and examiner-led academic decisions.

6 RESULTS AND VALIDATION

6.1 Research results

The supplied paper reports that renormalization reduced mean absolute score disturbance caused by missing measurements from 9.12 points under zero-imputation to 3.04 points on a 100-point scale across thirteen conditions, a factor of 3.00. Zero-imputation produced a negative mean score shift in every reported missingness condition, ranging from −3.88 to −22.61 points. Renormalization's signed shifts for individual sub-features stayed between −2.39 and +1.51 points. The effect was largest when an entire family was unavailable, because zero-imputation continued to allocate weight to values that had not been measured.

The pooled grade-band change rate reported by the paper was 65.8% under zero-imputation and 29.5% under renormalization. Losing the transcript family shifted the mean score by −22.61 under zero-imputation compared with −4.27 under renormalization; losing the engagement family shifted it by −20.79 compared with −1.55. Losing the acoustic family produced a positive mean shift of +5.84 under renormalization because the acoustic sub-features scored lower than other families in the simulated population. That positive shift is not an inconsistency: renormalization removes the unavailable family and averages the remaining measured families. It preserves evidence of availability, not a fixed expected score level.

Table 6.1. Missing-data ablation results reported by the supplied paper

Ablated signal	Renormalized mean ΔS	Zero-imputation mean ΔS	Renormalized grade changes	Zero-imputation grade changes
CNN engagement	−0.02	−10.41	35.6%	88.4%
Facial confidence	+0.02	−10.38	36.2%	88.3%
Pitch stability	+1.51	−4.28	34.4%	43.0%
HNR clarity	−2.39	−6.88	34.8%	67.6%
Articulation	+0.88	−4.70	30.0%	46.9%
Transcript sub-features	−0.72 to +0.80	−5.09 to −3.88	10.7–13.7%	38.8–51.0%
Engagement family	−1.55	−20.79	42.7%	99.7%
Acoustic family	+5.84	−15.87	60.0%	96.8%
Transcript family	−4.27	−22.61	48.3%	99.7%
Pooled mean absolute ΔS	3.04	9.12	29.5%	65.8%

6.2 Product and technical results

Table 6.1 presents the key missing-data values reported by the paper. Across the thirteen specified conditions, availability-aware renormalisation reduces mean absolute score disturbance and average grade-band changes relative to zero-imputation. The family-level results also identify where each input family has the greatest effect on the aggregate, helping prioritise calibration and subsequent evaluation.

Across the sampled conditions, zero-imputation preserved slightly higher average rank correlation, while availability-aware renormalisation yielded substantially smaller absolute score shifts and fewer grade-band changes. This comparison highlights the design priority of maintaining score calibration under missingness while keeping rank behaviour visible for future rubric-specific evaluation. Table 6.2 consolidates the reported sensitivity and correlated-degradation findings.

Table 6.2. Sensitivity and correlated-degradation results reported by the paper

Analysis	Reported result	Technical interpretation
±10% input perturbation	Maximum single-input movement 1.52 points; average by input no more than 1.03	Local sensitivity across the defined feature ranges
Engagement inputs	Mean absolute change 1.030 (CNN) and 1.027 (facial confidence)	Highest reported influence; prioritise assessment-construct alignment evaluation
Constant sweeps	Mean absolute score change at most 1.15; minimum rho 0.9924	Relative ordering stable within tested parameter ranges; calibration can align settings to the institutional rubric
Correlated degradation	2.82 points renormalization vs 9.09 zero-imputation; grade changes 26.3% vs 62.6%	Reported advantage ratio 3.23 under a synthetic latent-factor population
Rank ordering	Mean Spearman rho 0.896 renormalization vs 0.911 zero-imputation	Small rank-stability trade-off despite better absolute-score calibration

6.3 User or stakeholder validation

The supplied paper evaluates formula behaviour under controlled simulation conditions. A complementary examiner-led study can connect these software results to assessment practice by comparing pre-specified delivery subscales with independent rubric ratings from viva recordings collected under appropriate consent and oversight. A proposed next study would begin with at least 20 authentic recorded viva sessions

6.4 Integration results

The supplied paper describes invariance tests designed to keep LLM-generated criterion scores and diagnostic aggregates from changing the official mark. It also describes checks for capture-quality states and both scoring modes. This design reinforces modularity: advisory components can enrich examiner support while the scoring calculation remains governed by its permitted input schema.

For a robust integration review, test cases should vary every excluded signal over its permitted domain while holding the official inputs fixed. The output score and official feature audit should remain unchanged. A complementary test should mutate each permitted input and confirm that only the intended family score and final score are affected. Mode B should remain pending when T is absent, and the system should reject invalid T values rather than coercing them to zero or clipping an incorrect unit silently. These are recommended verification checks; only those explicitly described in the paper should be called paper-reported tests.

6.5 Summary of findings

Across the reported conditions, availability-aware renormalisation reduces average absolute-score disturbance and lowers the average rate of grade-band change. The correlated-degradation experiment reports the same direction of effect, highlighting the value of explicit availability handling when multiple signals depend on recording quality. The results also help prioritise calibration and external evaluation of the feature families with the greatest influence on the aggregate.

The findings support an engineering conclusion: assessment software benefits from distinguishing unavailable measurements from low observed values, making its input policy explicit and returning interpretable assessment metadata. A next study can extend this design evidence through examiner ratings, rubric alignment, recording diversity and subgroup analysis.

7 DISCUSSION

7.1 Interpretation of findings

The threefold reduction in mean absolute score disturbance is meaningful because it quantifies an error mode that could otherwise be hidden behind an apparently objective number. When a recording failure is converted to zero, the grade incorporates evidence about equipment and processing reliability rather than only evidence about the candidate. Renormalization reduces that direct penalty by redistributing weight over the measured families. Its advantage is strongest when an entire family is missing, which is plausible for a camera or transcription failure that affects several features simultaneously.

The results also show that renormalization is not score-neutral. It changes the relative weight of the surviving families, and the reported rank correlation is slightly lower than under zero-imputation. In other words, it aims to preserve a defensible absolute score under partial observability, not to preserve the exact ranking that would have been produced by a full vector. This trade-off should be stated in system documentation and examined with real examiners before the policy is accepted institutionally.

7.2 Comparison with literature baselines and existing solutions

Compared with learned missing-modality fusion in medical imaging [5], the Viva Evaluation scoring engine adopts a simpler deterministic rule rather than estimating missing feature values. This keeps the calculation directly inspectable and limits the assumptions introduced by learned reconstruction. The reported simulations favour this rule over zero-imputation on average for the tested ranges; broader comparisons can examine additional imputation and weighting strategies.

Compared with language-model-driven viva systems [12], [13], the scoring engine targets a complementary layer: consistent treatment of input availability beneath the interaction and content-assessment services. Advisory LLM services can provide prompts and feedback, while the official behavioural score is governed by its defined inputs. Compared with explainability techniques [14], [15], the feature ledger provides score provenance; attribution methods can complement this by analysing model influence within a validated use case.

7.3 Research contribution

The research contributes an explicit formalisation of availability-aware multimodal aggregation. It makes feature availability a first-class property of a scoring input and applies the same policy to individual sub-features and complete families. The resulting specification distinguishes a measured value from its availability metadata and gives the scoring engine a consistent way to aggregate the evidence selected for each assessment.

The work connects availability-aware aggregation with a clearly defined input policy. Official inputs are admitted through a versioned schema, and behavioural tests are designed to confirm that supporting diagnostic and advisory outputs remain separate from the mark. The reported experiments quantify score disturbance and grade-band changes across independent and correlated missingness scenarios, providing a software-behaviour foundation for continued assessment-focused evaluation.

7.4 Product contribution

For the broader Gradex system, the described component contributes a stable scoring contract between the feature-extraction pipeline and the reporting layer. Feature extraction may evolve, but any new value needs an explicit status, transformation and justification before it is admitted to grading. This reduces the chance that experimental outputs such as emotion labels, raw gaze values or semantic-model scores alter official behaviour unintentionally. The same contract supports an auditable record of which families were available when a score was returned.

The product contribution is a consistent scoring contract between feature extraction and examiner-facing reporting. The documented schema, feature-status ledger and test approach support consistent input handling, traceable provenance and separation of official marks from supporting diagnostic services. Versioned integration tests and operational monitoring can extend this foundation as the platform evolves.

7.5 Limitations and threats to validity

A key direction for continued evaluation is to connect controlled software-behaviour findings with assessment validity. The reported simulations compare aggregation policies under specified input ranges; representative viva recordings can extend this

work through score-agreement analysis with qualified examiners, threshold calibration and accessibility review.

The scoring policy uses configured engineering constants and deterministic normalisers. The paper reports that its tested parameter sweeps changed mean scores by at most 1.15 points on average, with a minimum Spearman correlation of 0.9924. These findings indicate relative-order stability within the selected ranges; subsequent calibration with representative recordings and an agreed rubric can further align the settings with institutional practice. The supporting model inventory can also be accompanied by checkpoint provenance and validation fixtures.

A reproducibility package should preserve the software version, simulation generator, random seeds, test outputs and configuration ledger alongside released results. The reference list follows IEEE numeric style; citation details should be aligned with the final source copies before institutional submission. Table 7.1 summarises the principal validation priorities and actions.

Table 7.1. Validation priorities and recommended actions

Validation area	Design consideration	Recommended action
Examiner alignment	Connects delivery indicators with the intended assessment construct	Independent raters; written rubric; reliability and agreement analyses
Real-recording coverage	Measures policy behaviour across representative recording conditions	Evaluate with real recordings and pre-specified degradation cases
Score interpretation	Separates delivery indicators from technical mastery	Report subscales separately and retain a technical rubric score
Calibration	Keeps configured thresholds aligned with the intended rubric	Calibrate using labelled data with external review
Accessibility and subgroup review	Considers effects of accent, disability, lighting and device quality	Subgroup analysis; accessibility routes; human review
Reproducibility package	Versioned artefacts support repeatable review of the reported experiments	Archive version, seed, generator, tests, data summaries and checksums

7.6 Implications

The first implication is that missing-data policy belongs in the system specification and test plan as a core scoring rule. Each score-affecting feature carries a value, availability state and provenance. The scoring pathway records assessment status when its configured eligibility criteria call for review. Technical separation helps maintain the boundary between behavioural feedback and content grading, while continued evaluation can align both dimensions with the institutional rubric.

For examiners, the score is presented with context to support review: family scores, available inputs, applied mode, assessment metadata and guidance that the aggregate concerns delivery rather than technical knowledge. For students, the institution can explain the intended use, route for human review and handling of sessions requiring further assessment. For developers, ongoing work can validate influential features, calibrate thresholds and test consistent processing across representative inputs. These implications align with risk-management guidance for AI systems [16], [17] and human-centred educational AI [18].

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The component's value proposition is a transparent scoring framework for multimodal viva assessment: it distinguishes availability metadata from measured performance, shows which feature families contributed to the score and records assessment status alongside the result. It is designed for examiner decision support, integrating video, audio, transcript and technical-assessment services into a reviewable workflow. Potential target users include university departments, certification providers and assessment-platform teams seeking consistent and auditable assessment workflows.

8.2 Market and competitive position

The component is positioned around transparent scoring governance and missing-data integrity. Its differentiator is a deterministic rule that can be inspected and tested, alongside modular video, audio, transcript and examiner-support services. Relevant alternatives include manual examiner review, complete-case handling, zero-imputation and learned missing-modality fusion. A future commercialisation phase can assess competitive position, institutional demand and sustainability through examiner consultation, stakeholder discovery and controlled pilot evaluation, followed by comparison against baseline workflows using pre-defined criteria.

8.3 Business or sustainability model

Potential delivery models include a university-hosted installation with annual support, licensing to an existing assessment platform, or a per-assessment service. Selection can follow an assessment of hosting, transcription, storage, compute, support, security and compliance costs under representative workloads. A sustainable model includes human review for sessions requiring follow-up, data minimisation, defined retention, access controls and deletion procedures.

8.4 Scalability intellectual property and compliance

Scalability planning should measure video throughput, transcription latency, storage, concurrent sessions and service-health monitoring under representative workloads. Before external deployment, the team should document code ownership, third-party model and dataset licences, a software bill of materials, security requirements, data

flows, retention rules and institutional approval responsibilities. Scoring changes should be versioned and reviewed before new features affect marks.

8.5 Contribution to the Sustainable Development Goals

The system has potential alignment with Sustainable Development Goal 4 (Quality Education) if it supports more consistent assessment while maintaining access and appropriate human review. A controlled study can evaluate successful completion, review outcomes, student appeals, accessibility, examiner understanding and subgroup patterns to determine where the system contributes to inclusive and equitable quality education.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis examined the Gradex Viva Evaluation scoring engine and its availability-aware aggregation policy for automated viva assessment. The component combines bounded measures of visual engagement, acoustic delivery and transcript fluency, while using availability metadata to determine the permitted evidence contributing to a score. It averages eligible sub-features within each family, renormalises the available family weights and records the configured output mode and assessment metadata.

The supplied paper reports a 20,000-vector simulation in which renormalisation reduced mean absolute score disturbance from 9.12 to 3.04 points and average grade-band change from 65.8% to 29.5% across thirteen missingness conditions. The paper also reports a larger relative advantage under correlated degradation and limited sensitivity across the tested parameter sweeps. These outcomes support the engineering case for availability-aware aggregation in the specified scenarios. A complementary evaluation using recorded viva sessions, examiner rubrics and diverse recording conditions can extend the evidence to educational interpretation and stakeholder use.

9.2 Achievement of objectives

Objective 1 was addressed by documenting the permitted families and bounded feature mappings defined for the scoring engine, version 1.1. Objective 2 was addressed through family means and renormalised weights across available families. Objective 3 was addressed by documenting the versioned feature schema, input boundary, capture-quality checks and output modes. Objective 4 was addressed by analysing the score-disturbance, grade-change, perturbation, constant-sweep and correlated-degradation results reported in the paper. Objective 5 was addressed by setting out a future examiner-led validation plan.

The next development cycle can strengthen repeatability by packaging the scoring version, test fixtures, simulation generator, random seed, test outputs and configuration ledger alongside release artefacts. This will make future updates easier to review and help ensure that changes in scoring rules are evaluated consistently.

9.3 Individual research and product contributions

The research contribution is a deterministic, auditable treatment of partial observability for multimodal scoring. The product contribution is a defined boundary between the official behavioural score and diagnostic, emotional and language-model-derived signals, supported by assessment metadata and examiner-controlled technical assessment. The reported simulation provides a quantitative comparison with zero-imputation across its specified feature space. Together, the scoring policy and implementation architecture create a clear foundation for further assessment-focused evaluation.

The next evaluation stage can examine how the most influential visual-engagement indicators align with the intended assessment construct across languages, recording conditions and student subgroups. These analyses can inform calibration and deployment decisions while preserving examiner oversight and keeping technical knowledge assessment distinct from delivery indicators.

9.4 Future directions

First, package the scoring-engine version, environment lockfile, seeded simulation script, aggregate outputs and test logs as a reproducibility set. Second, conduct an ethics-approved study with a written rubric, independent examiners, reliability analysis and pre-specified metrics. Third, calibrate influential normalisers using labelled data and analyse group-wise outcomes. Fourth, define coverage criteria, record excluded features and test representative input conditions. Fifth, evaluate camera, microphone and transcript variation across accents and recording environments. Sixth, study examiner and student understanding of the score. Maintain human judgement as the final authority and use the resulting evidence to guide consequential deployment decisions.

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GLOSSARY

Selected definitions used in this thesis:

Availability mask - A representation indicating which feature values are observed and permitted for scoring.

Bounded normalizer - A deterministic transformation that maps a raw feature to a configured interval, here [0, 1].

Feature-validation ledger - A record of each candidate signal’s measurement status and whether it is permitted, diagnostic, excluded or awaiting validation.

Missing modality - An unavailable input stream or family of features, distinct from a poor value that has actually been measured.

Renormalization - Reallocation of nominal weights across the subset of feature families that remain available.

Zero-imputation - Replacing an unavailable feature with numeric zero before aggregation.

APPENDICES

The appendices contain a concise scoring-policy algorithm, a validation checklist and a transparent disclosure of drafting support. They supplement the explanation in the main chapters and do not replace it.

Appendix A - Scoring policy and validation protocol

A.1 Reference scoring algorithm

1. Evaluate capture integrity against the configured face and lip-motion checks; record the assessment status.
2. Read the versioned permitted-feature fields and availability metadata.
3. Map measured values through deterministic normalisers into [0, 1].
4. Average available sub-features within each family.
5. Apply the configured output status when no family is available for scoring.
6. Otherwise calculate the weighted mean after renormalising available family weights.
7. Mode A returns the behavioural score and grade band; Mode B uses an external technical score T in [0, 10] and awaits examiner publication.
8. Return the score, mode, grade band, availability metadata and relevant gate information.
9. Record the scoring version and feature-ledger status; keep advisory and diagnostic outputs separate from official inputs.

A.2 Validation checklist for a controlled study

- Obtain ethics approval and informed consent before collecting real viva recordings.
- Use separate rubric scores for delivery and technical content, with multiple qualified raters where feasible.
- Log recording conditions, language background and capture status for later analysis.
- Report inter-rater reliability, correlation, absolute error, intervals and grade-band results.
- Review accessibility and subgroup outcomes, uncertainty, exclusions and appeal routes.

- Archive the code version, seed, environment, test outputs and aggregate tables for repeatable review.

Appendix B - Academic integrity and AI use disclosure

Generative artificial intelligence (AI) disclosure. ChatGPT (OpenAI) supported formatting comparison, proofreading and consistency checking; refinement of the scoring-policy wording; incorporation of the supplied Viva Evaluation technology inventory; and editing of selected implementation, discussion, commercialisation and conclusion passages. Representative workflows included comparison with the supplied university template, structured synthesis of the backend, frontend, video, audio, transcript, LLM and scoring modules, and checks of headings, pagination, table/caption consistency and the 50-page limit. Document verification included rendering the Word file to PDF and reviewing the rendered pages. AI-assisted revisions affected selected wording in Chapters 1 and 5-9, preliminary navigation lists and this disclosure. The candidate remains responsible for final technical accuracy, references, individual-contribution attribution, official supervisor details and compliance with the module academic-integrity policy.